

Reconstruction of Boosted di-jets from di-Higgs

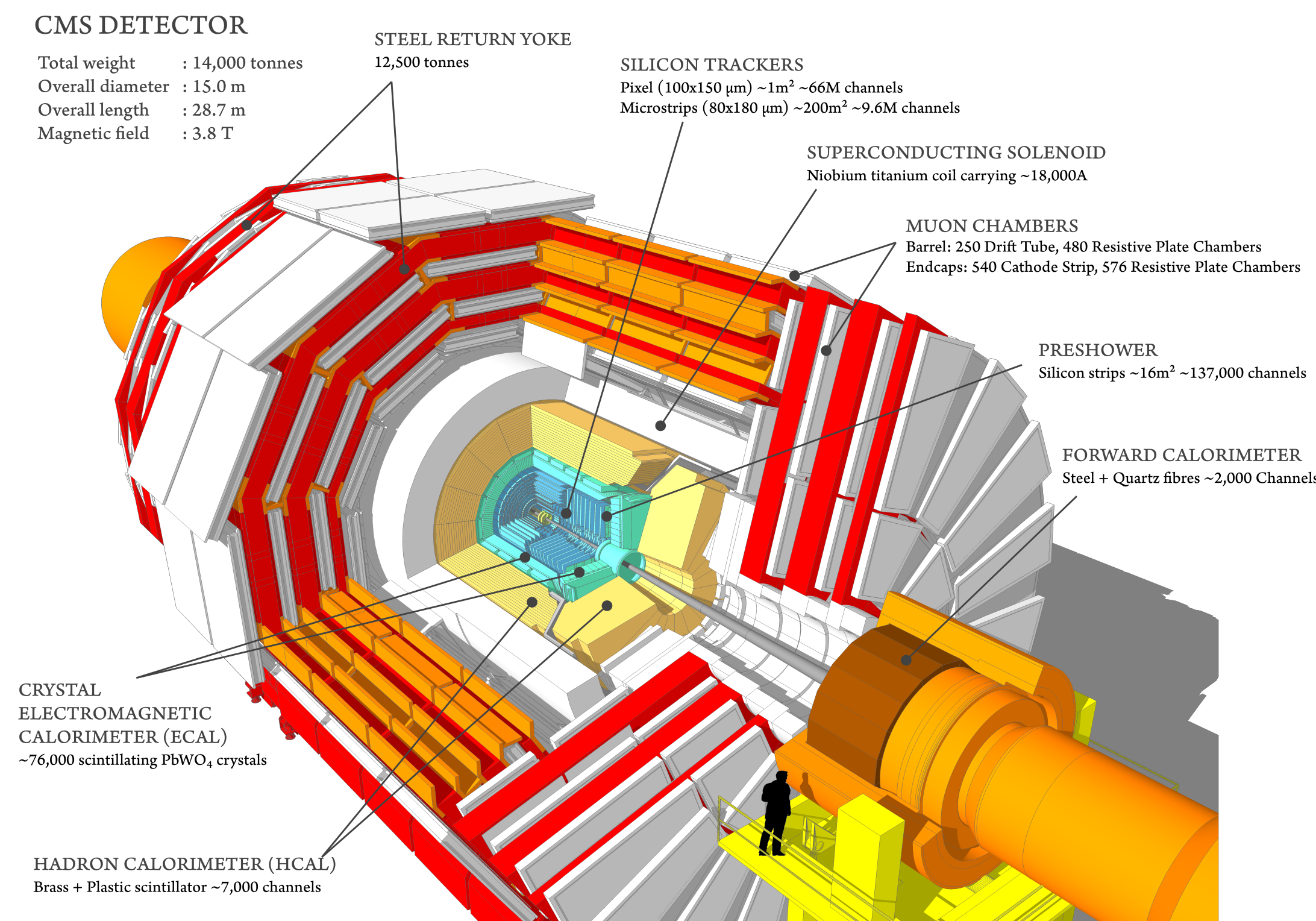
Boson Decay into Four Bottom Quarks

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Introduction

- The Compact Muon Solenoid (CMS) experiment is a general purpose experiment at the Large Hadron Collider (LHC) at CERN.
- The goal of the experiment is to engage in various topics in the expansion of physics such as exploring the Standard Model, or dark matter.

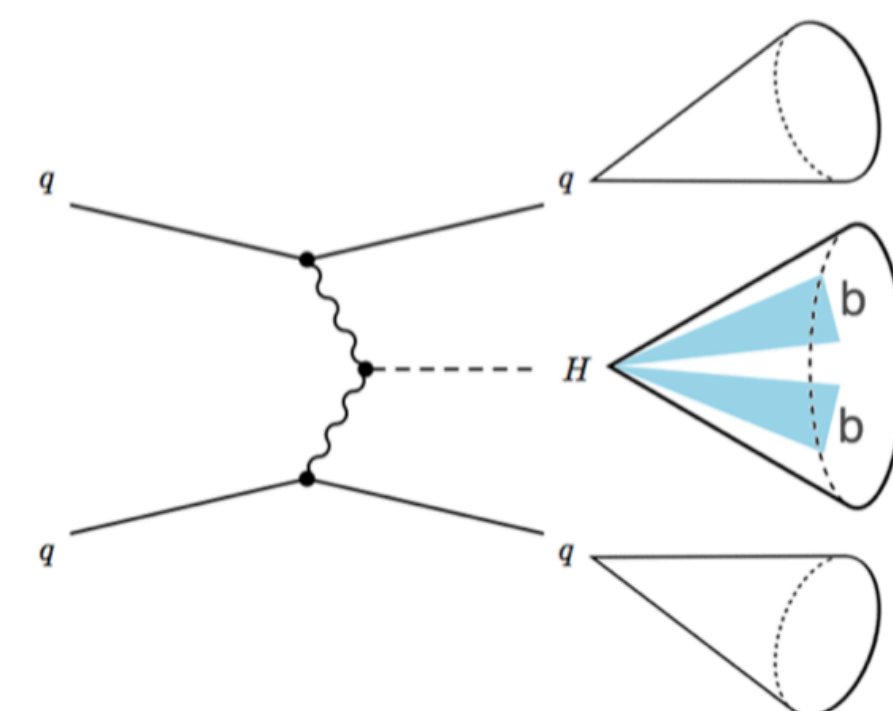


Cutaway diagram of CMS

The Higgs Boson

- The Higgs boson is a fundamental particle predicted by the Standard Model, and discovered at LHC in 2012
- The Higgs field, that gives mass to everything in the universe, is a manifestation of the Higgs Boson.
- The Higgs boson has self-coupling properties related to its mass that can lead to the infrequent production of two Higgs bosons, or di-Higgs boson particles.

Higgs produced through vector boson fusion (VBF), decaying into b-quark jets



Purpose

- Furthering our understanding of the Higgs boson and its properties can lead us to a greater understanding of the stability of the universe, of physics beyond the Standard Model, etc.
- Exploring di-Higgs boson decay processes in collisions at the LHC could illuminate the existence of new physics phenomena.



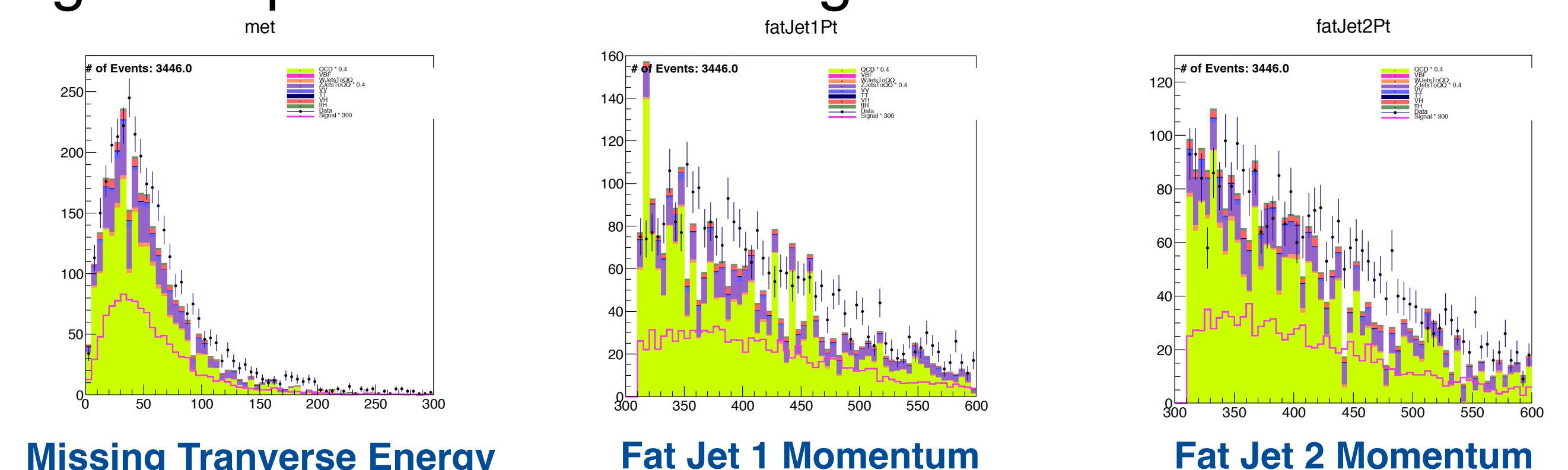
Bird's eye view of Compact Muon Solenoid at the Large Hadron Collider at CERN

Materials/Methods

- Using data from Run 2 (2016-2018) proton-proton collisions at 13 TeV and Monte Carlo simulations, I reconstructed plots for the jet mass, missing transverse energy (met), HH transverse momentum and mass, etc.
- Simulated background processes and signal were constructed through the use of generators, jet identification algorithms, or estimation utilizing dedicated data samples enriched in these processes.
- Selections are applied to the background processes as a large amount of extra particles are ejected in the collision

Results

- We construct our histograms, making selections upon variables such as fat jet momentum, PNet Xbb tagger, etc.
- The simulated signal in this region shows a peak, while the background processes is a falling distribution.



Missing Transverse Energy

Fat Jet 1 Momentum

Fat Jet 2 Momentum

Fat Jet 1 Mass

Fat Jet 2 Mass

Di-Higgs Momentum

Di-Higgs Mass

Conclusion

Constructing these histograms has provided a good understanding of the composition of the background regions that make up our data. We see that QCD and ZJetsToQQ make up a majority of the process. Moving forward due to our Monte Carlo statistical limitations, we would isolate the samples enriched with these dominant background sample processes individually to extract the shapes from these instead of using MC simulations.

Acknowledgments

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